

Netflix Content Analysis

Data Analytics Project | SQL (MySQL)

Prepared by: Welcome Molefe | Tool: MySQL Workbench | 2026

1. Introduction

This project demonstrates end-to-end data analytics using SQL on a Netflix titles dataset. The objective was to clean, explore, and analyse the dataset to extract meaningful insights about Netflix content distribution, popular directors, and audience targeting through ratings.

2. Dataset Overview

Attribute	Detail
Dataset	Netflix Titles
Tool Used	MySQL Workbench 8.0
Total Records	100 rows
Total Columns	show_id, type, title, director, country, release_year, rating
Source Format	CSV (imported via Table Data Import Wizard)

3. Data Exploration

Before cleaning, the dataset was explored to understand its structure and identify issues. The following missing values were identified:

Field	Missing Values	% of Dataset
Country	41	41%
Director	39	39%
Rating	0	0%
Type	0	0%
Title	0	0%

Missing values were represented as blank strings (") in the dataset, not true SQL NULLs, as confirmed by checking the raw data in Excel prior to import.

4. Data Cleaning

4.1 Duplicate Check

A duplicate check was performed using show_id as the primary identifier:

```
SELECT show_id, COUNT(*)
FROM netflix_titles
GROUP BY show_id
HAVING COUNT(*) > 1;
```

Result: Empty set — no duplicate records found.

4.2 Handling Missing Values

Given that 41% of country values and 39% of director values were missing, dropping these rows would have resulted in a loss of significant data. Following the 'Flag as Missing' strategy, blank values were replaced with 'Unknown':

```
SET SQL_SAFE_UPDATES = 0;
UPDATE netflix_titles SET country = 'Unknown' WHERE country = '';
UPDATE netflix_titles SET director = 'Unknown' WHERE director = '';
SET SQL_SAFE_UPDATES = 1;
```

Verification query confirmed all blanks were resolved:

Field	Missing After Cleaning
Country	0
Director	0
Rating	0
Type	0
Title	0

5. Data Analysis

5.1 Content Type Distribution

Type	Total	Percentage
Movie	55	55%
TV Show	45	45%

Insight: Movies make up 55% of the dataset while TV Shows account for 45%, indicating a relatively balanced content library with a slight preference for Movies.

5.2 Top 10 Countries by Content

Rank	Country	Total Titles
1	United States	17
2	Japan	13
3	India	6
4	United Kingdom	5
5	Nigeria	2
6	South Africa	1
7	United States, Ghana, Burkina Faso, United Kingdom...	1
8	Germany, Czech Republic	1
9	France, United States	1
10	Turkey	1

Insight: The United States leads with 17 titles, followed by Japan with 13. Notably, some records contain multiple countries in a single field, representing co-productions, which may affect country-level accuracy. Unknown values (41 records) were excluded from this analysis.

5.3 Top 10 Directors by Title Count

Rank	Director	Total Titles
1	Toshiya Shinohara	4
2	Masahiko Murata	3
3	Delhiprasad Deenadayalan	2
4	Hajime Kamegaki	2
5	Julien Ledercq	1
6	Mike Flanagan	1

Rank	Director	Total Titles
7	Stephen Kijak	1
8	Haile Gerima	1
9	Andy Devonshire	1
10	Kongkiat Komesiri	1

Insight: Japanese directors dominate the top spots, with Toshiya Shinohara directing 4 titles and Masahiko Murata directing 3, consistent with Japan's strong content representation in the dataset.

5.4 Rating Distribution

Rating	Description	Total
TV-MA	Mature Audiences (18+)	32
TV-14	Parents Strongly Cautioned (14+)	17
TV-PG	Parental Guidance Suggested	17
TV-Y7	Children 7+	10
PG-13	Parents Cautioned (13+)	8
PG	Parental Guidance	6
TV-Y	All Children	5
R	Restricted (17+)	3
TV-G	General Audience	2

Insight: TV-MA is the dominant rating with 32 titles — nearly a third of the dataset. Combined, mature content ratings (TV-MA, TV-14, R) account for 52 of the 100 titles, indicating Netflix's primary focus on adult audiences.

5.5 Content by Release Year

Release Year	Total Titles
2021	53
2020	8
2019	3
2018	4
2017	2

Release Year	Total Titles
2015 and earlier	30 (spread across years back to 1975)

Insight: 2021 dominates with 53 out of 100 titles, suggesting the sample was heavily drawn from recent content. The dataset also contains titles going as far back as 1975, showing Netflix carries classic content alongside new releases.

6. Key Findings

- Netflix content in this sample is split 55% Movies and 45% TV Shows.
- The United States and Japan are the top content-producing countries.
- Japanese directors lead in title count, aligned with Japan's strong country representation.
- Over half of all titles carry mature content ratings (TV-MA, TV-14, R), targeting adult audiences.
- 2021 accounts for 53% of titles, indicating the sample is heavily skewed toward recent content.
- 41% of country records and 39% of director records had missing values, retained and flagged as 'Unknown'.

7. Limitations

- The dataset contains only 100 rows, which limits the generalisability of findings to Netflix's full library.
- Some country fields contain multiple countries (co-productions), which inflates unique country counts.
- The heavy skew toward 2021 titles may not reflect true historical content distribution.
- Missing values (41 countries, 39 directors) were retained as 'Unknown' — conclusions involving these fields carry reduced accuracy.

8. Conclusion

This project successfully demonstrated a complete SQL-based data analytics workflow: data import, exploration, cleaning, and analysis. Using MySQL Workbench, meaningful insights were extracted about Netflix content distribution, audience targeting through ratings, and regional content production. The project highlights the importance of data cleaning decisions — particularly the choice to retain rather than drop missing values — and demonstrates how SQL's GROUP BY, COUNT, ORDER BY, and CASE WHEN constructs can be used to answer real business questions.

github.com/Melcomex | portfolio-ashen-five-83.vercel.app